

Confidence Interval Explained

When we make an estimate in statistics, whether it is a **summary statistic** or a **test statistic**, there is always uncertainty around that estimate because the number is based on a sample of the population you are studying.

The **confidence interval** is the range of values that you expect your estimate to fall between a certain percentage of the time if you run your experiment again or re-sample the population in the same way. The **confidence level** is the percentage of times you expect to reproduce an estimate between the upper and lower bounds of the confidence interval, and is set by the alpha value.

In short:

- A confidence interval is the mean of your estimate plus and minus the variation in that estimate.
- This is the range of values you expect your estimate to fall between if you redo your test, within a certain level of confidence.

Confidence, in statistics, is another way to describe probability. For example, if you construct a confidence interval with a 95% confidence level, you are confident that 95 out of 100 times the estimate will fall between the upper and lower values specified by the confidence interval.

Your desired confidence level is usually one minus the alpha (α) value you used in your statistical test; (where α is known as your level of significance):

$$\text{Confidence level} = 1 - \alpha$$

So, if you use an alpha value of $p < 0.05$ for statistical significance, then your confidence level would be $1 - 0.05 = 0.95$, or 95%.